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1 Part A: Detailed Design Documentation

1.1 Task 1: Evaluation of SDLC Models

1.1.1 Introduction

Software Development Life Cycle (SDLC) gives a systematic way of doing innovation in information systems. In the case of PASS FIRST TIME Driving School system, it is consistent to adopt appropriate SDLC model in order to implement it successfully. This section compares two SDLC models that can be used in this project(Cormen, 2022).

1.1.2 Waterfall Model

The waterfall model is a conventional, sequential model which has to have each phase can be passed off before progressing to the next stage. The model has different stages requiring analysis, system design, implementation, testing, deployment, and maintenance(Sedgewick Wayne, 2021).

The Waterfall Model aspects:

Waterfall model is a linear model. The first requirement is to compile all the requirements and documented(Bryant O'Hallaron, 2021). Then, system design is made according to these requirements. After design approval, coders write the code. Testing occurs when the coding has been done. And lastly, the system is implemented and serviced.

Advantages: The driving school system can use the Waterfall model having a number of advantages. The structure is not complicated and coherent to all team members. Each phase has definite deliverables and deadlines, and therefore the project management is easy(Russell Norvig, 2021). Every stage is well documented and this is beneficial in the later maintenance. The model suits the situation when there are no uncertainties or differences in the requirements. For the driving school, the most necessary things such as making reservations, storage of student data, and the oversight of instructors are clear and there are not apt to be altered greatly.

Disadvantages: The Waterfall model however has some limitations. After one stage is followed, transition. It is not easy and cheap to go back and make changes. In case requirements were poorly understood first, this can only be made plain in the course of testing resulting in delays. Users cannot see the working system till end of the project and would result in dissatisfaction in case of anticipations do not correspond with the final product. In the case of the driving school, when employees understand that they should add such options as payment history or student status report, add them would be challenging.

PASS FIRST TIME: Suitability:

Waterfall model is the reasonably appropriate one with regard to this project. The driving school system is small to medium-sized having well-defined requirements. The technical complexity is comparatively low, and entails simple database manipulation and user interfaces. However, the model may be inflexible and this might be an issue should the driving school desire to introduce features during development(Suleyman, 2023).

1.1.3 Agile Model

Agile model is a form of iterative model which provides working software in small portions increments. The process of development takes place in short successive cycles often referred to as sprints that last two to four weeks(Kurose Ross, 2024).

Aspects of the Agile Model:

Agile is concentrated on communication and adaptability. The requirements are collected on a continuous basis throughout the project. Minor part of the system is developed, constructed and validated each sprint(Suleyman, 2023). After every iteration users give comments and this can enable developers to make changes the system accordingly. The team has frequent meetings where progress is discussed challenges.

Advantages:

The driving school project can experience much benefits of agile(Stallings Brown, 2024). The school can see working entails providing feedback and early on in the process to make sure the end system is satisfactory to them needs. Between sprints, changes are easily added. Risk is reduced because issues are detected and resolved very fast. In case the driving school makes the decision midway that they would like to have automatic reminder messages of their lessons, this can be included in the next sprint. The involvement of the users during the development would result in increased satisfaction with the final product(Martin, 2023).

Disadvantages:

Agile also has drawbacks. The driving is at a loss of time commitment to the model regularly scheduled feedback meetings between school staff. failing which they are not well documented upfront, new team members can have a more difficult time getting familiar with the project. Costs and timelines can be not as predictable as Waterfall. In case of a small driving school with few staff time, frequent use of it may be problematic(Vaswani et al., 2017). Moreover, in case of scope of the project as the number of requests per sprint kept growing, the project may end up being out of budget timeline.

PASS FIRST TIME: Suitability:

The PASS FIRST TIME driving school system requires the Agile model that is very suitable. Surface development The project size is fitting to iterative development(Cormen, 2022). Basic features like lesson It is possible to deliver booking and add later and more advanced features such as reporting. The driving school is able to test every module and ask to be improved. Given that the brief mentions such as progress of students, payments can be added in the future, Agiles has integrative style would be ideal. Until it can implement advanced functionality, the school can begin with basic functionality further provisions are being constructed.

1.1.4 Recommendation

In the case of PASS FIRST TIME driving school system, the Agile model will be more appropriate. It would be of benefit to the project to develop it in a slow but sure progression manner faster and improved according to end-user feedbacks. The school has the opportunity to start utilizing the. The features of booking and student management and the reports and analysis tools are designed during following sprints. This will minimize risk and provide the ultimate system really fulfills the operational requirements of the school(Sedgewick Wayne, 2021). Nevertheless, a concise scope management is necessary to avoid the endless features additions that will slow down its completion.

1.2 Task 2: Information Gathering Methods

1.2.1 Interview based Requirements Gathering

The interviews with stakeholders are crucial in the context of knowing the business requirements of the Pass 1st Time Driving School. The interviews with the manager can be used to determine the goal, including enhancing the efficiency of the scheduling process, decreasing the amount of administrative tasks, and boosting student retention (Bryant O'Hallaron, 2021). Teachers and office personnel offer insights on the day to day issues such as keeping track of the student progress and bookings. The interviews are used to make the new system consistent with business growth strategy and operation requirements (Russell and Norvig, 2021).

1.2.2 Feeding Back the User with Surveys

With surveys, numerous students can leave their feedback about booking, payments, and interactions with instructors in a short amount of time and anonymously (Russell and Norvig, 2021). This feedback assists in determining features required e.g. online booking and lesson reminders, which enhances user satisfaction. Surveys offer a wide view of the user when used alongside interviews and this will ensure that the system fits the business needs and those of the student (Bryant and O'Hallaron, 2021).

1.3 Task 3: Data Flow Diagrams and Flowchart

1.3.1 Data Flow Diagrams

The Data Flow Diagrams (DFDs) depict the flow of the information in the PASS FIRST TIME driving school system. They contribute to the visualization of processes, data stores as well as external entities they were not focused on technical implementation.

Context Diagram (Level 0):

The Context Diagram depicts the boundary of the system and external parties. The main Students, Instructors, Office Staff, and the Driving Test Centre are the external entities. Lessons are booked and confirmed to students. Teachers are handing in lesson records and receive schedules. The system contains office personnel who create reports. The Driving Test Centre receives information on test booking and returns the results to the system(Suleyman, 2023).

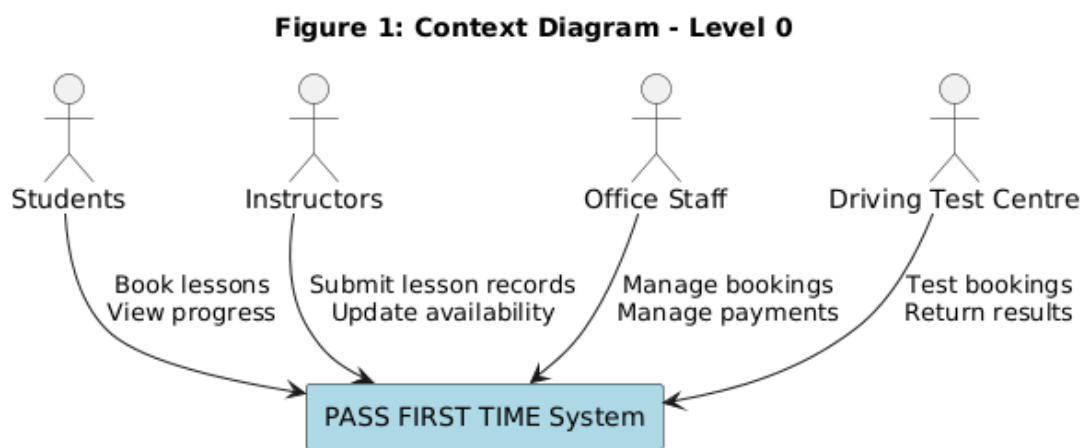


Figure 1: Pass first time Context Diagram level Zero

Level 1 DFD:

Level 1 diagram divides the system into major processes: Manage Student Registration, Manage Lesson Booking, Manage Instructor Allocation, Process Test Bookings, and Generate Reports. These processes and data are linked by the data flows such as stores as Student Database, Lesson Records, and Instructor Information.

Figure 2: DFD Level 1

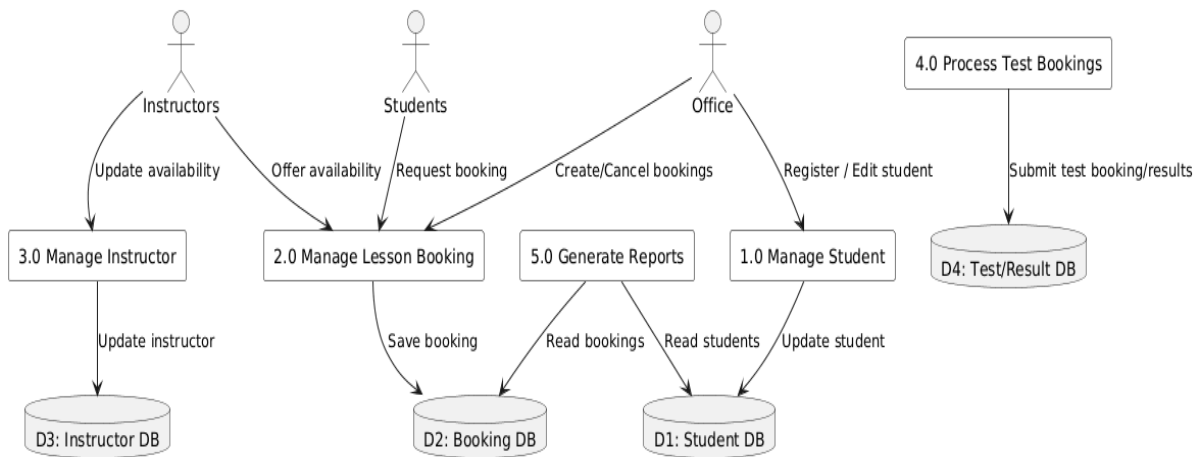


Figure 2: Pass first time Context Diagram level One

Level 2 DFD (Process of Lesson booking):

Level 2 diagram gives a close-up impression of the process of lesson booking and displays sub-processes such as Check Instructor Availability, Validate Student Details, Calculate Fee, and Reserve Booking(Suleyman, 2023).

Figure 3: DFD Level 2 (Lesson Booking)

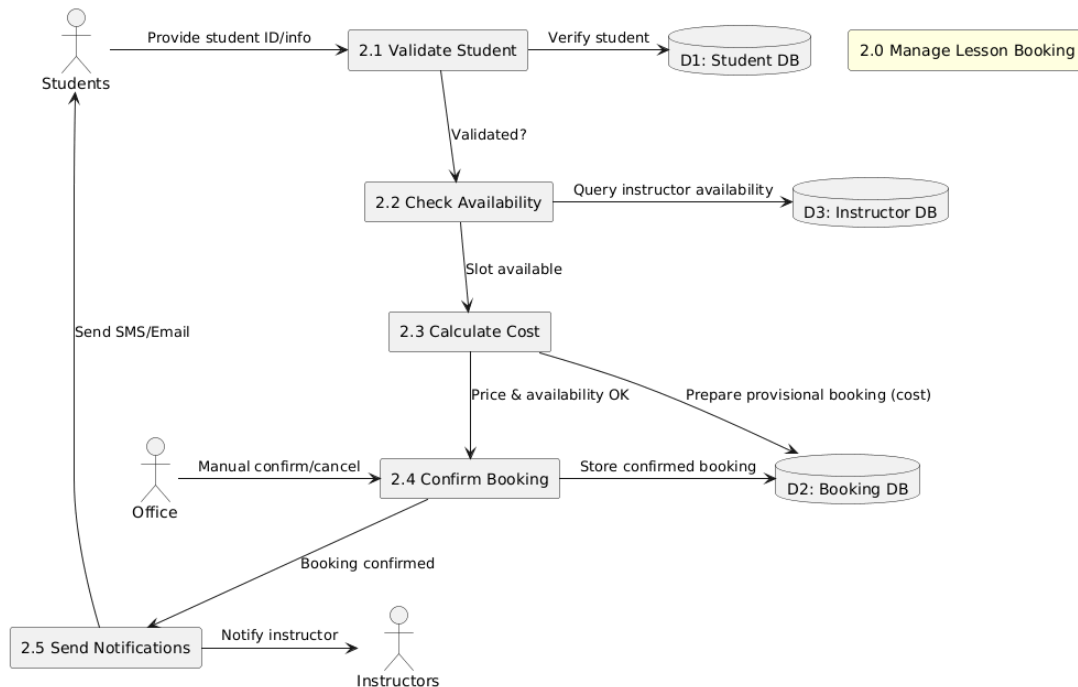


Figure 3: Pass first time Context Diagram level Two

1.3.2 Flowchart

The flowchart shows how booking of the lesson happens. It starts when a student makes a lesson booking request. The system determines the existence of the student in the system database. Otherwise, the registration of the students is necessary. Then the availability of instructors is checked at the date and time required. Provided it is available, the cost of the lesson is calculated based on lesson type. The reservation is registered and stored in the database. Finally, both the student and the instructor are given confirmation(Suleyman, 2023).

Figure 4: Flowchart - Lesson Booking Process

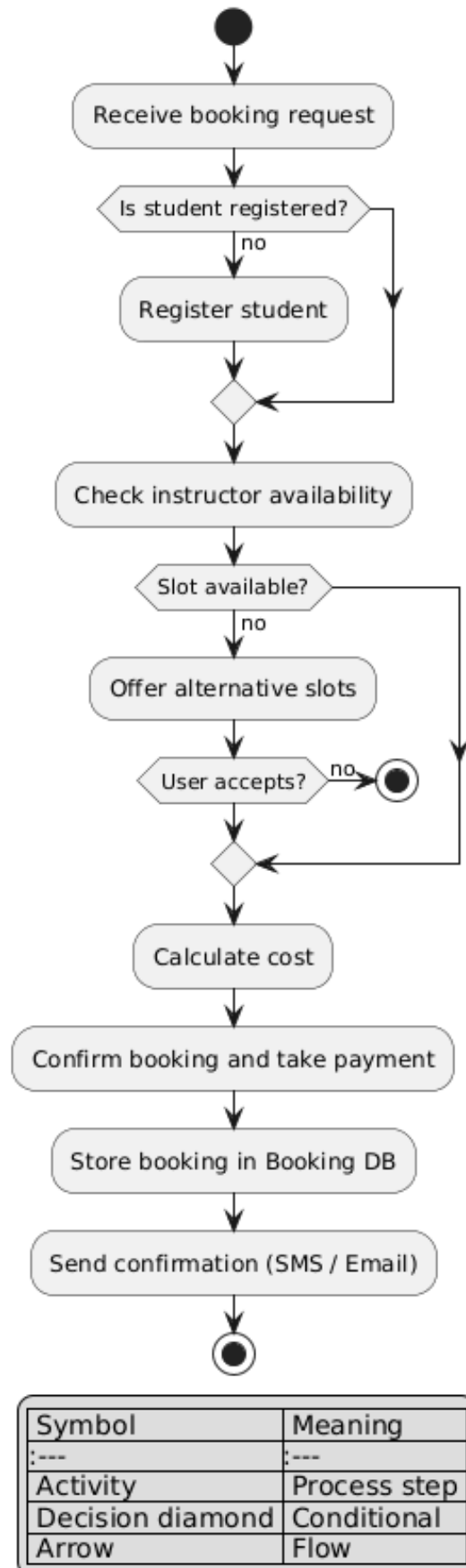


Figure 4: Lesson booking Process flowchart

1.4 Task 4: User Interface Design

1.4.1 Data Entry Forms

The PASS FIRST TIME system must have a number of user friendly forms to help with efficient data entry. All the forms must be straightforward, user-friendly, and reduce mistakes in data entries at a minimum level.

Student Registration Form:

This type gathers personal information of the students such as full name, date of birth, address, phone number, email, driving license number (where applicable). The form uses Most fields have text boxes, date picker on the birth date and dropdown menu on the title selection. Asterisks are placed against mandatory fields. Checking Email validation checks format is good and there are no non-digit phone numbers(Kurose Ross, 2024).

STUDENT REGISTRATION FORM

PASS FIRST TIME Driving School

Fields marked with * are required

Personal Information

Title *

Select

Full Name *

Enter first name, middle name, and last name

Date of Birth *

dd-mm-yyyy

Gender

Select

Nationality

Enter nationality

UK Driving License Number

e.g., SMITH751234AB9CD

Optional - leave blank if not obtained

Theory Test Status

Select status

Theory Test Pass Date

dd-mm-yyyy

Contact Details

Primary Phone Number *

e.g., 07123 456789

Alternative Phone Number

Optional

Landline or Mobile

Email Address *

student.email@example.com

Full Address *

Street address, building number

City/Town *

Enter city or town

Postcode *

e.g., B1 1AA

SUBMIT

CLEAR

CANCEL

Figure 5: Student Registration form For Pass First Time System

Lesson Booking Form:

This type of form permits the booking of new lessons. The user gets to choose the name of the student in a dropdown list, select lesson type (introduction, standard, pass plus or driving test), select date and time with calendar and time pickers, lesson duration, and select an instructor. The cost is automatically calculated and automatically set in the form. A notes Section permits the introduction of special instructions(Kurose Ross, 2024).

Instructor Management Form:

This form is used by office staff to insert or update the information on an instructor such as their name, contact type of employment (full-time and part-time), qualification, and availability schedule. The instructor availability is displayed under the form of a weekly calendar view.

1.4.2 Hardware and Software Requirements

Hardware Requirement: The system will require a server computer with minimum 8GB RAM and 500GB memory to store the database. Computers of clients (at least 3) having 4GB Indoor office personnel and instructors will have access to the system using RAM. The internet should be reliable for real-time updates. Printing schedule and print report printer.

Software Requirements: Database management system MySQL or Storing all the data with the use of Microsoft SQL server. Apache or IIS web server applications to host the application(Stallings Brown, 2024). Current internet browsers (Chrome, Firefox, or Edge) on the client computers. Server Operating system: Windows 10/11 or Linux. Development tools Programming languages such as PHP, Python, or Java can be used to create the application interface. The user interface is focused on simplicity and efficiency and saves on time on staff training and at the same time maintaining accuracy of data by means of validation and clear layouts.

2 Part B: System Evaluation

2.1 Task 1: Factors Affecting System Success and Failure

2.1.1 Introduction

The effectiveness of PASS FIRST TIME driving school system relies on a number factors. By knowing of the possible risks and challenges, it becomes possible to come up with strategies to counter the same guarantee successful implementation and sustainability(Stallings Brown, 2024).

2.1.2 Risk Connected with Meeting Time and Budget

Some of the project risks are modifications in specifications, technical complications, and employee attrition, which might slow down the delivery period and raise the expenses (Martin, 2023). The following risks can be mitigated by using the project milestones, frequent progress review, and contingency budget of 15-20 percent. Agile methodology offers the opportunity to make flexible changes without significant impact on the schedule, whereas clear communication assists in keeping realistic deadlines (Kurose Ross, 2024).

2.1.3 Costs and Time Constraints

System success is influenced by financial and time constraints. Programmer salaries, software, hardware and training are covered under development costs (Kurose and Ross, 2024). Hurrying to meet deadlines during peak seasons means sacrificing quality and satisfaction with the user. The priorities include prioritizing key functionality, open-source operating systems, cost-effective hardware, and implementing in stages to distribute the expenses (Stallings and Brown, 2024).

2.1.4 Organizational Policies

Organizational policies influence the implementation of systems (Russell and Norvig, 2021). Privacy and access policies should be in line with such regulations as GDPR. Integration requires the desire to embrace the digital records; a combination of both the paper and the system lowers efficiency. Clearly defined policies on the use of the systems, backups, password, and authorized changes, as well as training of staff before going live, are required (Stallings and Brown, 2024).

2.1.5 Staff Skills and System Maintenance

System adoption is influenced by organizational policies (Russell and Norvig, 2021). Protection of data, access control and compliance to GDPR are essential. To be integrated effectively, it is necessary to transfer the manual to the digital records. There should be clear instructions regarding the use, backups, passwords, authorized modifications, and training of staff (Stallings and Brown, 2024).

2.1.6 Change Management and User Involvement

Resistance to change often leads to system failure since the new systems can be perceived by the employees as a threat or additional work. A sense of ownership can be achieved through involvement, and successful adoption results in effective change management, through clear benefit, training, engagement, and leadership support (Russell and Norvig, 2021; Bryant and O'Hallaron, 2021).

2.1.7 Conclusion

Risk management is the key to system success, involving risks of technical, financial, organizational, and human factors. Proper planning, sensible budgets and schedules, good integrity, support to management, friendly design, extensive training and support maintenance establish the circumstances under which the PASS FIRST TIME system will be a success enhancing the operations of the driving school (Russell Norvig, 2021).

2.2 Task 2: Design Evaluation and Future Improvements

2.2.1 Comparison to User Requirements

The PASS FIRST TIME driving school system also eliminates the use of paper records instead of using secure computerized database where students and lesson management is easy. It also deals with reservations, checks on the availability of instructors, eliminates cases of booking two reservations, student tracking, test administration, and reports. The role-based, simple interface is efficient in terms of its data entry and communication, which makes staff members concentrate on customer service (Bryant O'Hallaron, 2021).

2.2.2 Future Developments and possible Improvements

Efficiency and satisfaction can be enhanced through the addition of automated email or SMS reminders to reduce the number of missed lessons to the system. A personalized booking platform with open slots, up-to-date lesson history, and the payment options would enable the students and decrease the workload in the office (Russell and Norvig, 2021). Visual dashboards used to track the progress would enable the instructors to give improved feedback and inspire students. Financial management would be enhanced by incorporating the payment processing, invoicing and reporting. Mobile applications would enable real time updates in lesson records, which is time-saving and more precise. Staffing, marketing, and seasonal planning could be determined using historical data analytics. The system would also be integrated with the external systems such as the DVSA database that would minimize the administrative work. Such improvements are to be made in stages according to priority, budget and user response as per the Agile ideals.

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